SAYAN GHOSH

MSc in Data Science and AI, RKMVERI, Belur

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SUMMARY

Data Science and AI postgraduate with hands-on experience in Machine Learning, Deep Learning, Reinforcement Learning. Skilled in developing AI-driven solutions, predictive models, and distributed systems using Python, PyTorch, and Scikit-Learn. Strong analytical and problem-solving abilities, backed by a solid foundation in mathematics, statistics, and data science.

EDUCATION

Singur Mahamaya High School
Madhyamik – 653/700 (93.3%), 2020
HS – 477/500 (95.4%), 2022

Presidency University , Kolkata
BSc (Honours) in Mathematics, 2022–25
CGPA – 9.09

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Data Science and Artificial Intelligence, 2025–27
CGPA – 8.64 (till Sem 2)

SKILLS

- **Languages:** Python, C Programming, SQL, LaTeX, R Programming
- **Frameworks:** Scikit-Learn, NumPy, Pandas, Streamlit, PyTorch, Matplotlib
- **Operating Systems:** Windows, Ubuntu, Linux
- **Tools:** Git/GitHub, MS Office, MS Excel

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning, Natural Language Processing, Emerging Technologies : RAG, Blockchain
- Probability and Statistics, Linear Algebra, Graph Theory, Time Series and Survival Analysis
- Distributed Computing, Data Structures and Algorithms, DBMS

ACHIEVEMENTS

- DST Inspire Scholarship (Top 1% of board performers, Govt. of India)
- SVMCM Scholarship (Govt. of West Bengal)
- JAM 2025 (Mathematiics) - Qualified
- NMMSE Scholarship

CERTIFICATES

- Online Foundation Course in Mathematics (MTTS) 2023
- Crest Crypto Summer School (TCG CREST)

INTEREST

- Solving Mathematical Problems
- Cryptography
- Playing and Analyzing Cricket

PROJECTS

Distributed Credit Card Fraud Detection using Apache Spark & H2O

RKMVERI Feb 2026 – Apr 2026

Developed a distributed machine learning pipeline for large-scale credit card fraud detection using Apache Spark, Sparkling Water, and H2O Random Forest. Implemented data preprocessing, distributed partitioning, class imbalance handling, threshold optimization, and model evaluation using fraud-focused metrics including Precision, Recall, F1-Score, Balanced Accuracy, and ROC-AUC. Designed the system to support scalable execution across Spark clusters for efficient processing of large transaction datasets.

Hybrid Energy-Based and Normalizing Flow Models for Trustworthy LLM Output Evaluation

RKMVERI Feb 2026 – Apr 2026

Developed a research-oriented framework for evaluating the trustworthiness of Large Language Model outputs by combining Energy-Based Models with Normalizing Flow architectures. Used BERT embeddings for feature extraction and integrated energy-based confidence estimation with probabilistic uncertainty modeling to detect hallucinated and unreliable responses. Evaluated multiple flow architectures and designed a hybrid trust-scoring framework for reliable confidence assessment of AI-generated content.

AI Guided Reinforcement Learning Framework for Automated Web GUI Testing

RKMVERI Oct 2025 – Dec 2025

Designed a reinforcement learning framework for automated GUI testing by transferring policies learned in Grid World environments to Selenium-driven web applications. Implemented Q-Learning and Deep Q-Network (DQN) approaches for intelligent GUI navigation, automated login testing, interaction with dynamic web elements, and error-state detection. Demonstrated reinforcement learning as an adaptive alternative to traditional rule-based GUI testing workflows.

Smart City Traffic Light Control Using Reinforcement Learning

RKMVERI Oct 2025 – Dec 2025

Developed an intelligent traffic management system using Reinforcement Learning (RL) to optimize traffic signal operations in urban environments. Designed and trained an RL agent to dynamically adjust signal timings based on real-time traffic density, vehicle queues, and waiting times. Implemented simulation-based testing to evaluate system performance, achieving improved traffic flow, reduced congestion, and lower average vehicle waiting times compared to traditional fixed-time traffic control methods.

Cricket Score Prediction System using Regression Based Machine Learning Models

RKMVERI Sep 2025 – Nov 2025

Built a machine learning-based prediction system to estimate cricket match scores using historical match statistics and team performance metrics.Implements multiple regression models and comparative evaluation techniques to generate accurate score forecasts and support sports analytics applications.